(iii) The equation (4.29), $T = mg\cos\theta + \frac{mv^2}{r}$ highlights that in sections A and D of the circle, $\left(\text{for } -\frac{\pi}{2} < \theta < \frac{\pi}{2};\ \cos\theta \text{ is positive}\right)$, the term mg cosθ is always greater than zero. Hence the tension cannot vanish even when the velocity vanishes.

(iv) The equation (4.29), $\frac{mv^2}{r} = T - mg\cos\theta$; further highlights that in sections B and C of the circle, $\left(\text{for } \frac{\pi}{2} < \theta < \frac{3\pi}{2};\ \cos\theta \text{ is negative}\right)$, the second term is always greater than zero. Hence velocity cannot vanish, even when the tension vanishes.

These points are to be kept in mind while solving problems related to motion in vertical circle.

To start with let us consider only two positions, say the lowest point 1 and the highest point 2 as shown in Figure 4.15 for further analysis. Let the velocity of the body at the lowest point 1 be $\vec{v}_1$, at the highest point 2 be $\vec{v}_2$ and $\vec{v}$ at any other point. The direction of velocity is tangential to the circular path at all points. Let $\vec{T}_1$ be the tension in the string at the lowest point and $\vec{T}_2$ be the tension at the highest point and $\vec{T}$ be the tension at any other point. Tension at each point acts towards the centre. The tensions and velocities at these two points can be found by applying the law of conservation of energy.

For the lowest point (1)

When the body is at the lowest point 1, the gravitational force $m\vec{g}$ which acts on the body (vertically downwards) and another one is the tension $\vec{T}_1$ acting vertically upwards, i.e. towards the centre. From the equation (4.29), we get

$$T_1 - mg = \frac{mv_1^2}{r} \qquad (4.30)$$

$$T_1 = \frac{mv_1^2}{r} + mg \qquad (4.31)$$

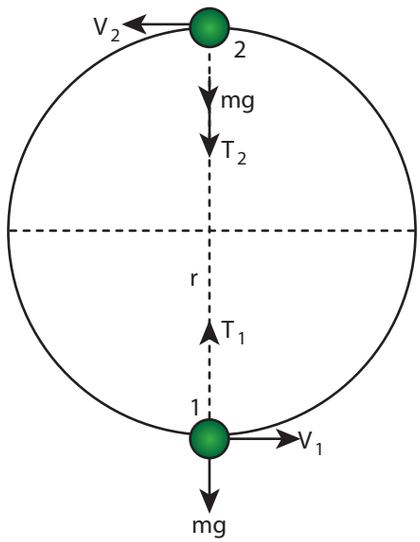


Figure 4.15 Motion in vertical circle shown for lowest and highest points

For the highest point (2)

At the highest point 2, both the gravitational force $m\vec{g}$ on the body and the tension $\vec{T}_2$ act downwards, i.e. towards the centre again.

$$T_2 + mg = \frac{mv_2^2}{r} \qquad (4.32)$$

$$T_2 = \frac{mv_2^2}{r} - mg \qquad (4.33)$$

From equations (4.31) and (4.33), it is understood that $T_1 > T_2$. The difference in tension $T_1 - T_2$ is obtained by subtracting equation (4.33) from equation (4.31).

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